Gabriel Serrano Rojas

Full-Stack Software Engineer | **Portfolio:** gabrielserranorojas.com
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Education

Universidad de Costa Rica Mar. 2023 – Dec. 2026 (Expected)
Bachelor of Science in Software Engineering | GPA: 8.9/10 *San José, Costa Rica*
- **Coursework:** Data Structures & Algorithms, Design Patterns, Advanced Databases, Web Development

Universidad de Costa Rica Mar. 2021 – Present
Bachelor of Science in Physics (concurrent degree) *San José, Costa Rica*

Experience

Full-Stack Engineer Mar. 2024 – Present
MangaKaCR Bookstore *San José, Costa Rica*
- Built a C (.NET 9) clean-architecture POS/inventory system (ASP.NET Core API, Blazor, PostgreSQL), cutting costs 78%
- Designed a REST API with 20 endpoint groups and JWT auth, consumed through a Kiota-generated typed client
- Built sales analytics dashboards: 8-dimension drill-down rankings, period/YoY trends, layaway-aware KPIs
- Created 6 payment flows; delivered invoicing, layaways, cashback, audited voids and shift cash-reconciliation reports
- Fixed lost-update races with PostgreSQL advisory locks; built live cross-tab sync with in-process pub/sub
- Wrote 916 tests (87 Testcontainers integration tests); built CI/CD with CodeQL, SonarQube and Docker
- Integrated the tax authority's (Hacienda) API; automated a daily catalog PDF sync to Google Drive
- Developed the Wix store (∼3,000 monthly users, ∼$2,000 monthly revenue); building a Next.js/React storefront

Student Research Assistant Feb. 2026 – Present
Universidad Estatal a Distancia / Universidad de Costa Rica *San José, Costa Rica*
- Digitized biomedical questionnaires into an offline HTML/CSS/JS app that keeps patient data in local CSVs
- Built CSV merge and schema-restore tools to reconcile diverging questionnaires and ensure data quality
- Supervised migrating 600+ paper questionnaires to the system, now used by 10 other research assistants

Projects

ThemePark@UCR | *C#, .NET, Blazor, SQL Server, Entra ID, Azure* Sep. 2025 – Dec. 2025
- Owned platform-wide Entra ID sign-in and role-based permissions, enforced on every API endpoint and page
- Set up deployment to Azure App Service with environment-specific config and Bearer-token validation
- Served as Scrum Master of one of the course's teams and sat on the project-management and Gitflow committees

LASCE Research Lab Platform | *Next.js, React, TypeScript, Prisma, PostgreSQL* Aug. 2026 – Present
- Designed the PostgreSQL schema (Prisma + SQLAlchemy) and publications module for UCR's space weather lab
- Developed a reusable in-place CMS and news module with role-gated REST API routes and MinIO image uploads
- Built the admin dashboard with authentication/authorization, tested with Vitest, Storybook and Playwright

Virtual Microscope Administration Panel | *C#, .NET, Blazor, PostgreSQL, Electron* Mar. 2026 – Jul. 2026
- Replaced in-memory repositories with an EF Core + PostgreSQL persistence layer, mappings and migrations
- Developed the Blazor WebAssembly frontend with a typed REST API client, CRUD pages and paginated search

Distributed ASCII-Art Network | *C++, Sockets, HTTP, L2/L3 Switching* Mar. 2025 – Jul. 2025
- Developed the team's C++ HTTP client, fork server and drawing servers in a client–server architecture

Technical Skills

Languages: C#, SQL (PostgreSQL, SQL Server), C/C++, HTML/CS, STypeScript, JavaScript, Python
Web & Data: ASP.NET Core, EF Core,React, Blazor, Next.js, Node.js, RESTful APIs, Prisma, SQLAlchemy
Testing & DevOps: xUnit, Vitest, Playwright, Testcontainers, Git, GitHub Actions (CI/CD), Docker, SonarQube
Cloud: Azure (App Service, Entra ID), Google Cloud, Supabase | **Practices**: Agile/Scrum, Gitflow, clean architecture

Languages

Spanish: Native | **English**: C1 Advanced (EF SET certified)